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The Study

This analysis aimed to reproduce the results from Lépissier, Alice & Mildenberger, Matto. (2021). Unilateral climate policies can substantially reduce national carbon pollution. Climatic Change. 166. 10.1007/s10584-021-03111-2. The authors employed a synthetic control method (SCM) to estimate the causal effect of the UK's 2001 Climate Change Programme (CCP) on the country's carbon emissions. Their findings demonstrate that CCP reduced the UK's carbon emissions. CO2 emissions per capita were 9.8% lower in 2005 relative to what they would have been if the CCP had not been passed, demonstrating that unilateral climate policies can still reduce carbon emissions even in the absence of a binding global climate agreement.

The 2001 UK CCP is a complex reform that included a carbon tax on large-scale energy users, industry-negotiated exemptions from the tax for meeting reduction targets, and a voluntary emissions trading scheme. It was one of the first comprehensive climate reform packages passed globally, in advance of action by most other Organisation for Economic Co-operation and Development (OECD) countries. Importantly, there is a limitation to evaluating the post-policy enactment effect, as the EU emissions trading system (EU ETS) was enacted in 2005.

This positions the UK's CCP to be evaluated against a wide pool of donor countries by pre-treatment carbon emissions. (Megan has already done this)

Data

- data: all countries
- data_OECD: similar economic conditions
- data_OECD_HIC: + high income countries
- data_OECD_HIC_UMC: + upper middle countries

Setup

Lépissier & Mildenberger (2021) generated synthetic controls using an optimization algorithm to select donor countries to match pre-CCP emissions trajectories in the UK and the synthetic control as closely as possible. This algorithm generates the synthetic UK minimizing mean squre predication error (MSPE) of the pre-treatment outcome variable. The authors take the synthetic control route as it is less restrictive than a difference-in-difference strategy. The key assumptions of this method are:

1. SUTVA (stable unit treatment values assumption). There is no spillover of CCP enactment to other countries in the donor pool

2. Variables used to form the weights must have values for donor pool countries that are similar to those of the UK. This minimizes interpolation bias if the procedure interpoulates across regions with very different characteristics

Does the setup of the treatment satisfy these assumptions?

The authors selected the donor sample based on countries that had not passed any major climate legislation until the 2005 European Union's Emissions Trading Scheme (EU ETS). The UK was the only country with the CCP in 2001. *Outcome variable*: CO2 emissions per capita

Ensures that emissions data are on the same order of magnitude across pool- avoiding interpolation bias

The authors additionally rescale CO2 emissions per capita to a 1990 and 2000 baseline

Donor pool specifications

Donor pool consisted of 51 countries which were either OECD members or classified by the World Bank as upper middle–income or high-income countries at the time of treatment in 2001, that had a population greater than 250,000, and that did not have a carbon pricing policy in place.

The Work Bank classifies countries into income categories according to GNI per capita in US dollars. In fiscal year 2001, the World Bank classified high-income (HIC) countries as those with GNI per capita above 9265 US dollars, and upper middle–income (UMC) countries as those with GNI per capita in the 2996 US dollars to 9265 US$ range. "EN.ATM.CO2E.PC" (CO2 emissions per capita in metric tons).

CO2 emissions measured are those stemming from the burning of fossil fuels and the manufacture of cement (power sector).

Recreating UK Synthetic Control

We used the {tidysynth} package to generate our synthetic control UK, as compared to the authors' optimized algorithm method which involved a manual econometrics approach.

```
library(tidysynth) # for synthetic control
library(tidyverse)
library(patchwork) # for combining plots
library(here)
```

We sourced the original emissions datasets from 1. OECD countries, 2. high-income countries (OECD), and 3. high-income (OECD_HIC) and upper-middle-income countries (OECD_HIC_UMD).

▶ Code

First, we attempted to create a synthetic control that averaged CO2 emissions per capita across the pre-treatment time period by country.

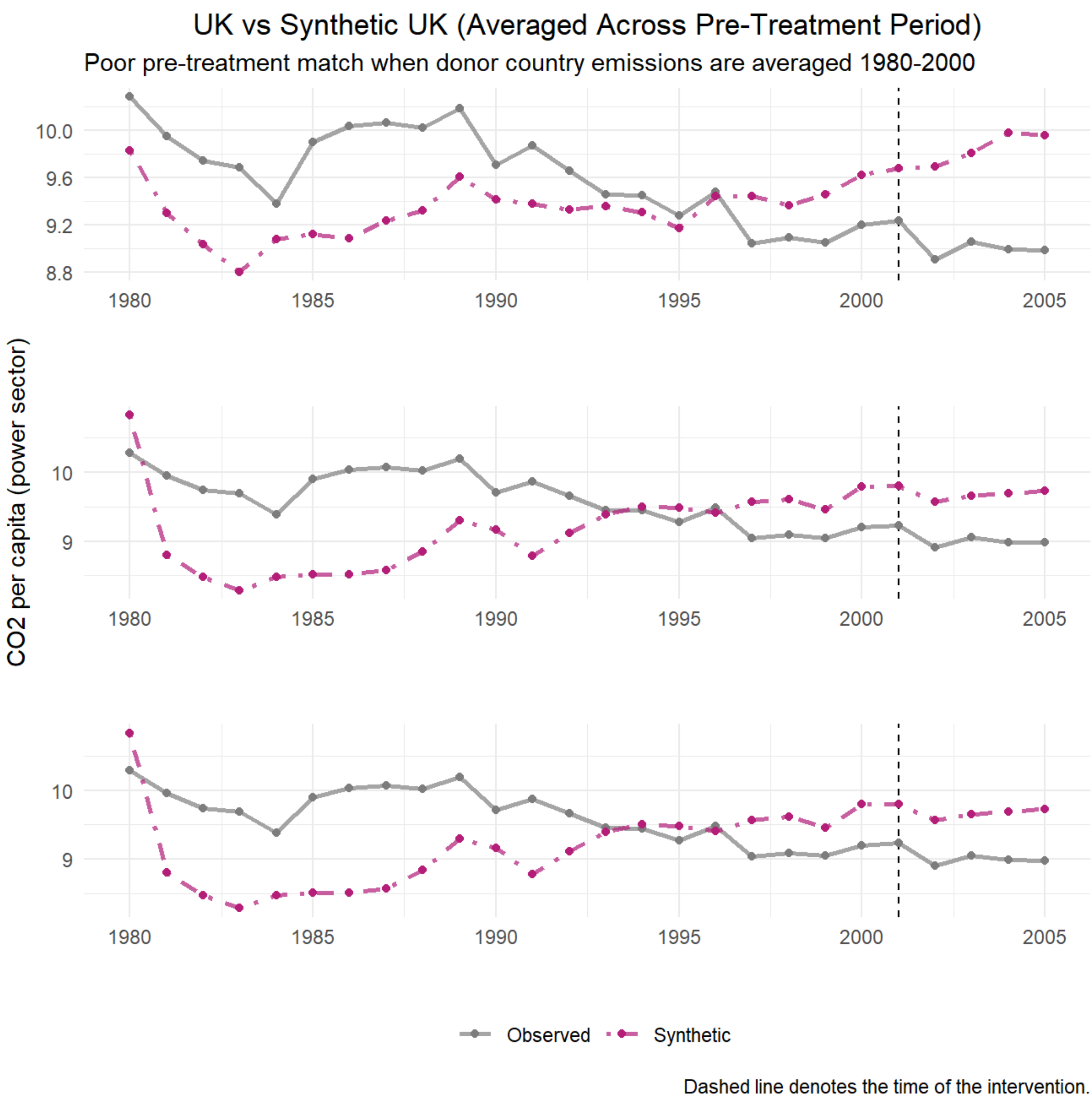
▶ Code

By plotting the trends of the resulting synthetic controls across time, we can see that this method results in a poor pre-treatment fit, which violates the assumption of the synthetic control method to make a meaningful causal effect estimation.

▶ Code

```
Warning: Using `size` aesthetic for lines was deprecated in ggplot2 3.4.0.
ℹ Please use `linewidth` instead.
ℹ The deprecated feature was likely used in the tidysynth package.
  Please report the issue to the authors.
```

▶ Code

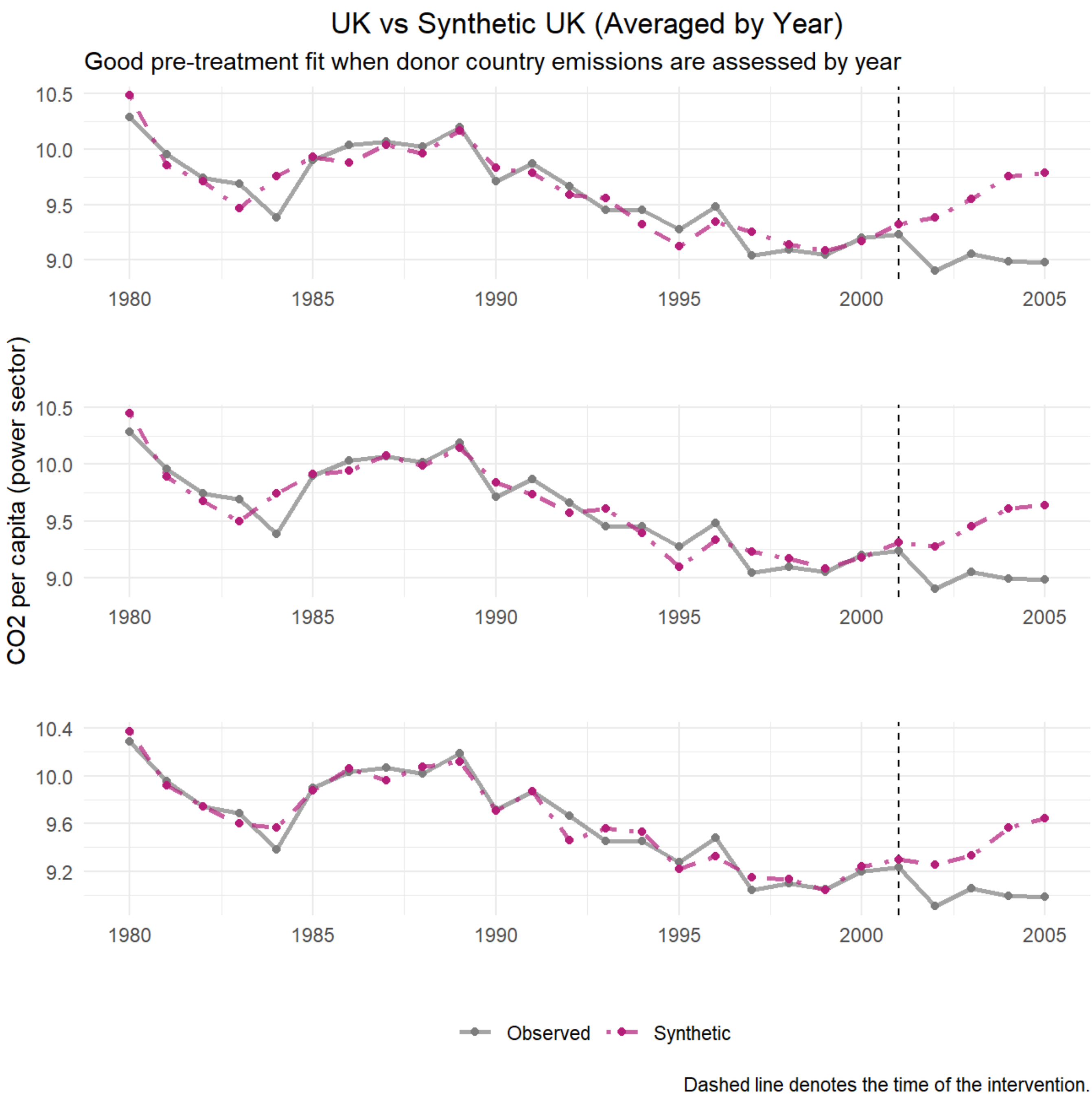


Due to this outcome, we generated synthetic outcomes based on yearly emissions by donor country.

▶ Code

Plotting the outcomes of synthetic control across all 3 donor pool options, we see a much better pre-treatment fit than the previous method of synthetic control.

▶ Code



Since the pre-treatment fit assumption is met in our synthetic control recreation, we go on to compare gaps between synthetic UK and observed UK emissions to evaluate treatment effect.

Plot Gaps (Treatment Effect)

▶ Code

```
# A tibble: 5 × 4
  time_unit real_y synth_y     gap
      <int>  <dbl>   <dbl>   <dbl>
1      2001   9.23    9.30 -0.0645
2      2002   8.90    9.25 -0.347
3      2003   9.05    9.33 -0.276
4      2004   8.99    9.57 -0.578
5      2005   8.98    9.64 -0.658
```

▶ Code

```
# A tibble: 11 × 2
   unit        weight
   <chr>        <dbl>
 1 Poland      0.182
 2 Italy       0.165
 3 Germany     0.116
 4 Mexico      0.112
 5 Austria     0.110
 6 Greece      0.0838
 7 Australia   0.0813
 8 Luxembourg  0.0652
 9 Switzerland 0.0319
10 Portugal    0.0194
11 New Zealand 0.0144
```

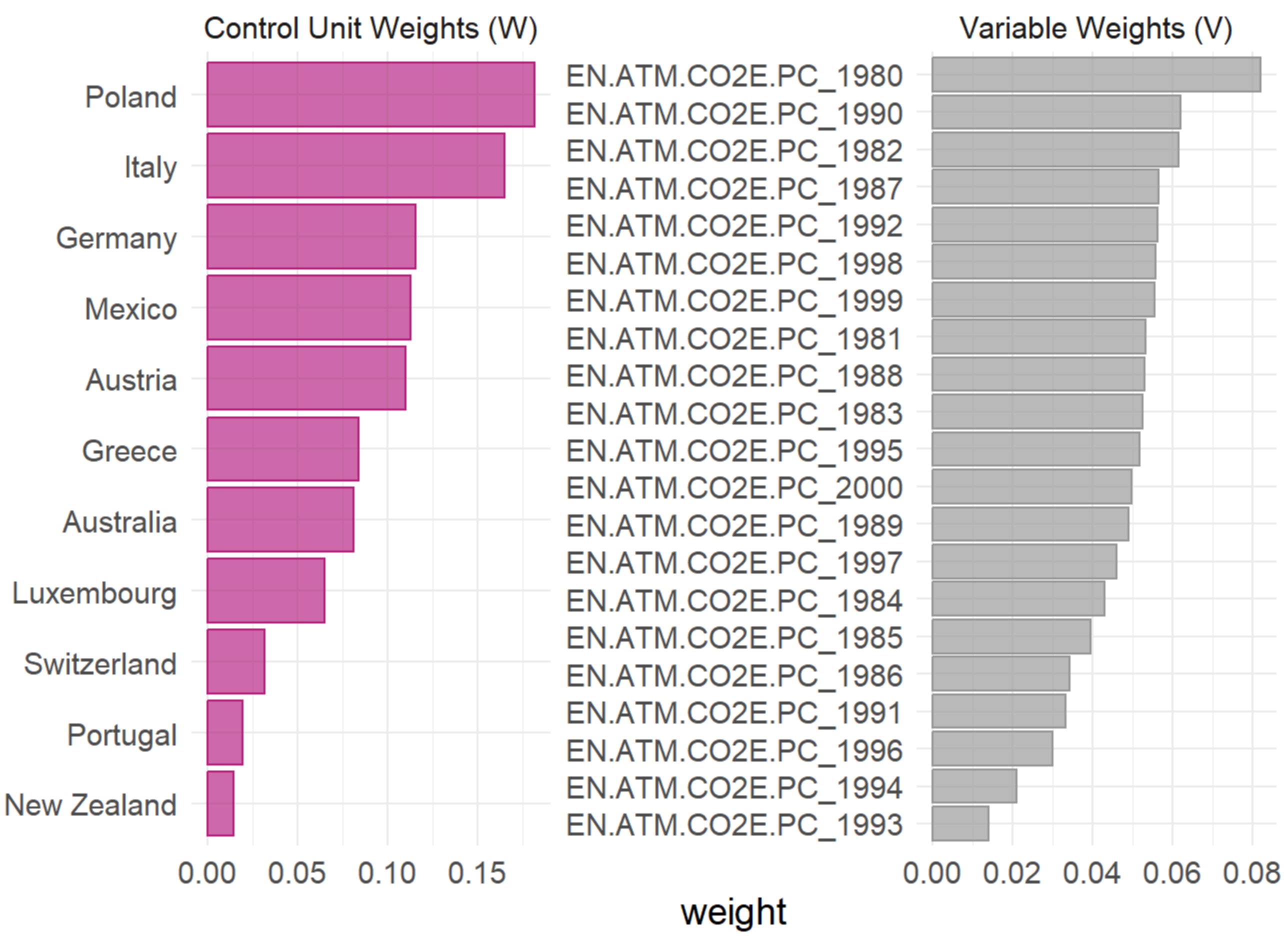
▶ Code

```
# A tibble: 11 × 2
   unit          weight
   <chr>          <dbl>
 1 Germany       0.172
 2 Mexico        0.136
 3 Austria       0.133
 4 Poland        0.126
 5 Macao         0.120
 6 Luxembourg    0.0654
 7 Australia     0.0639
 8 Hong Kong     0.0541
 9 United States 0.0507
10 Israel        0.0506
11 Italy         0.0103
```

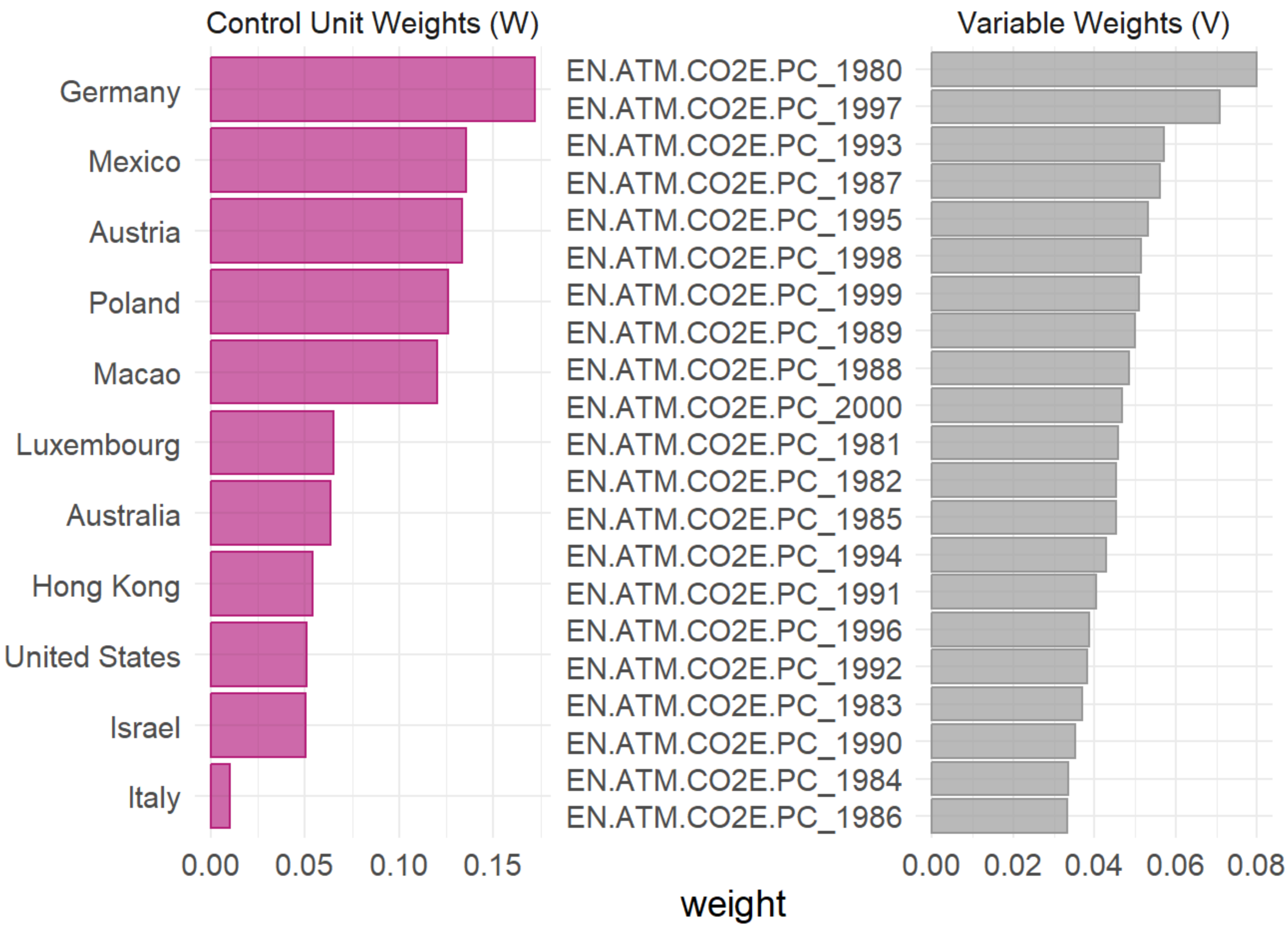
▶ Code

```
# A tibble: 9 × 2
  unit         weight
  <chr>         <dbl>
1 Libya        0.234
2 Germany      0.216
3 Macao        0.178
4 Poland       0.0967
5 Australia    0.0867
6 Hong Kong    0.0834
7 Luxembourg   0.0479
8 Saudi Arabia 0.0230
9 Cyprus       0.0107
```

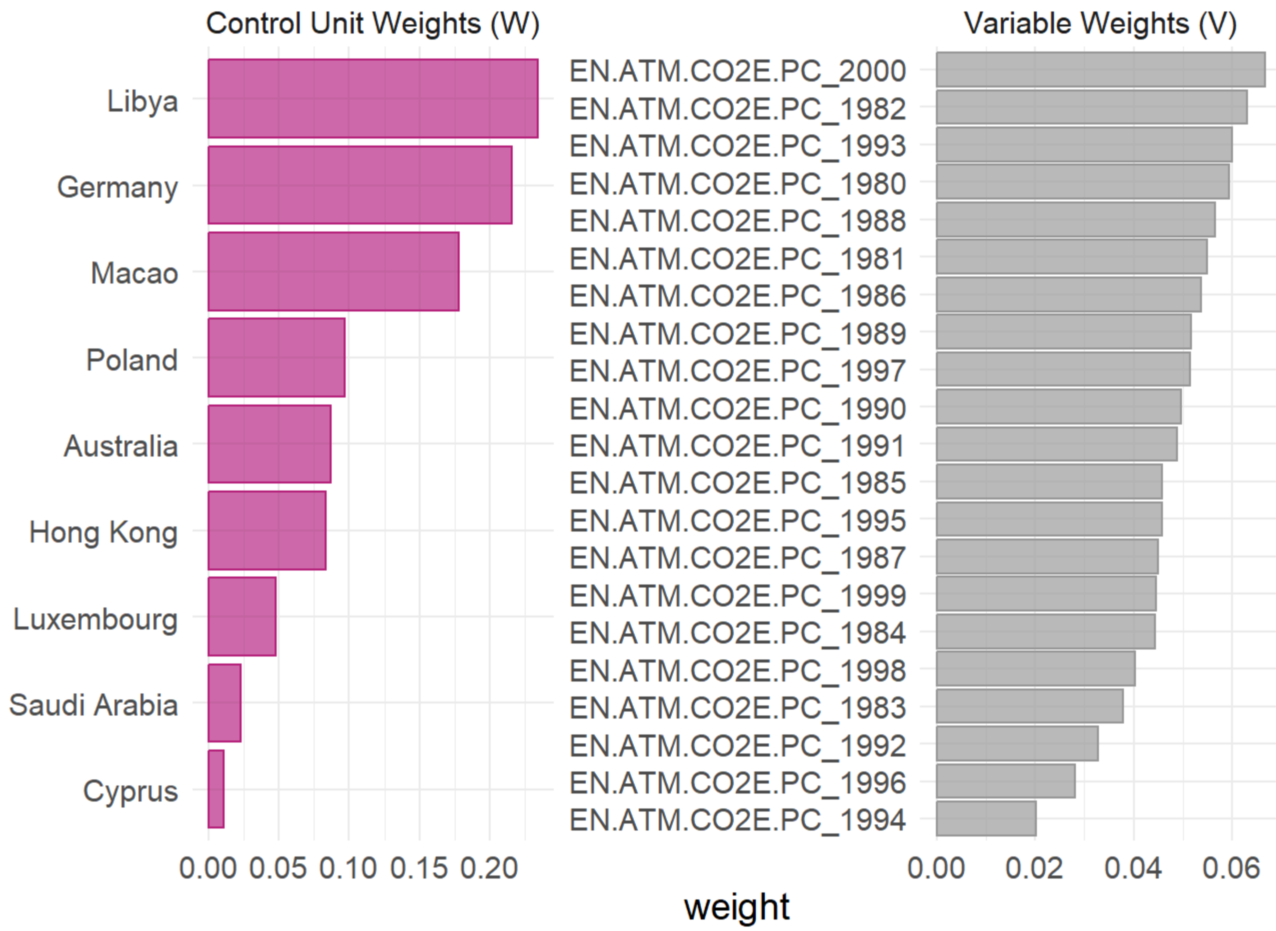
▶ Code



▶ Code


Control Unit Weights (W)
Germany
Mexico
Austria
Poland
Macao
Luxembourg
Australia
Hong Kong
United States
Israel
Italy
0.00 0.05 0.10 0.15
Variable Weights (V)
EN.ATM.CO2E.PC_1980
EN.ATM.CO2E.PC_1997
EN.ATM.CO2E.PC_1993
EN.ATM.CO2E.PC_1987
EN.ATM.CO2E.PC_1995
EN.ATM.CO2E.PC_1998
EN.ATM.CO2E.PC_1999
EN.ATM.CO2E.PC_1989
EN.ATM.CO2E.PC_1988
EN.ATM.CO2E.PC_2000
EN.ATM.CO2E.PC_1981
EN.ATM.CO2E.PC_1982
EN.ATM.CO2E.PC_1985
EN.ATM.CO2E.PC_1994
EN.ATM.CO2E.PC_1991
EN.ATM.CO2E.PC_1996
EN.ATM.CO2E.PC_1992
EN.ATM.CO2E.PC_1983
EN.ATM.CO2E.PC_1990
EN.ATM.CO2E.PC_1984
EN.ATM.CO2E.PC_1986
0.00 0.02 0.04 0.06 0.08
weight

▶ Code



MSPE Ratios

Comparing MSPE is an integral element in the orginal study, as minimizing differences ensures the best synthetic control outcome. The pre-treatment MSPE of the best fitted donor pool was 1.24 × 10−4. The MSPE should be as close to 0 as possible. Based on this, our best fitted synthetic pool is the high income and upper middle income pool.

```
scm_data_OECD %>%
  grab_significance() %>%
  filter(unit_name == "United Kingdom")
```

```
# A tibble: 1 × 8
  unit_name      type   pre_mspe post_mspe mspe_ratio  rank fishers_exact_pvalue
  <chr>          <chr>     <dbl>     <dbl>      <dbl> <int>                <dbl>
1 United Kingdom Treat…   0.0193     0.427       22.1     2               0.0870
# ℹ 1 more variable: z_score <dbl>
```

```
scm_data_OECD_HIC %>%
  grab_significance() %>%
  filter(unit_name == "United Kingdom")
```

```
# A tibble: 1 × 8
  unit_name      type   pre_mspe post_mspe mspe_ratio  rank fishers_exact_pvalue
  <chr>          <chr>     <dbl>     <dbl>      <dbl> <int>                <dbl>
1 United Kingdom Treat…   0.0175     0.276       15.8     7                0.212
# ℹ 1 more variable: z_score <dbl>
```

```
scm_data_OECD_HIC_UMC %>%
  grab_significance() %>%
  filter(unit_name == "United Kingdom")
```

```
# A tibble: 1 × 8
  unit_name      type   pre_mspe post_mspe mspe_ratio  rank fishers_exact_pvalue
  <chr>          <chr>     <dbl>     <dbl>      <dbl> <int>                <dbl>
1 United Kingdom Treat…  0.00786     0.241       30.6     3               0.0577
# ℹ 1 more variable: z_score <dbl>
```

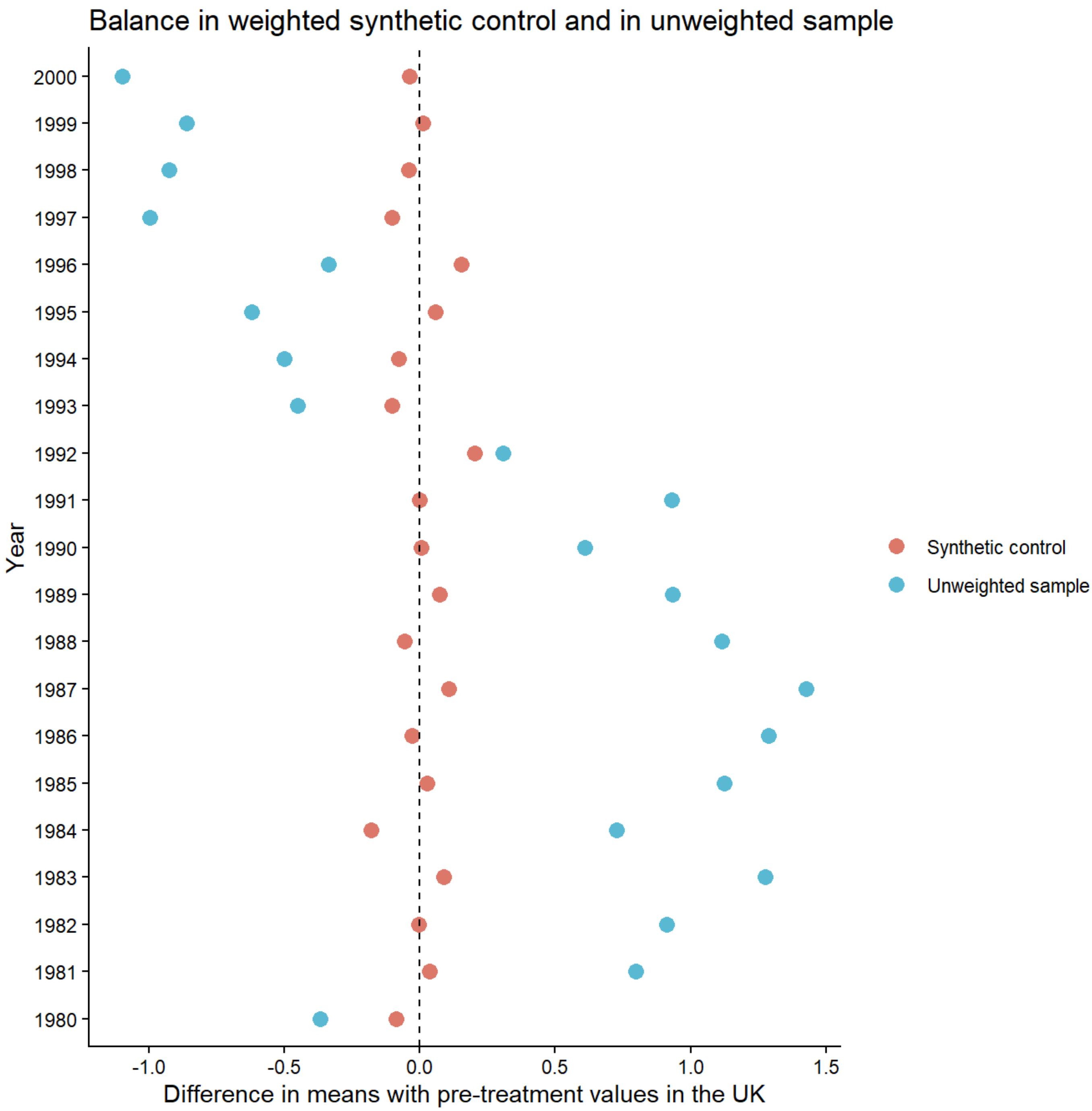
```
scm_data_OECD_HIC_UMC %>%
  grab_significance() %>%
  filter(pre_mspe < 5 * min(pre_mspe[type == "Treated"])) %>%
  arrange(desc(mspe_ratio))
```

```
# A tibble: 14 × 8
   unit_name      type  pre_mspe post_mspe mspe_ratio  rank fishers_exact_pvalue
   <chr>          <chr>    <dbl>     <dbl>      <dbl> <int>                <dbl>
 1 United Kingdom Trea…  0.00786   0.241       30.6       3               0.0577
 2 Argentina      Donor  0.00644   0.118       18.4       7               0.135
 3 Spain          Donor  0.0169    0.280       16.6       8               0.154
 4 Italy          Donor  0.00326   0.0450      13.8       9               0.173
 5 Greece         Donor  0.0275    0.352       12.8      10               0.192
 6 Austria        Donor  0.0358    0.325        9.07     11               0.212
 7 Macao          Donor  0.0223    0.118        5.28     14               0.269
 8 Mexico         Donor  0.0151    0.0699       4.62     18               0.346
 9 Japan          Donor  0.0253    0.0852       3.37     21               0.404
10 Cyprus         Donor  0.0246    0.0776       3.16     23               0.442
11 Brazil         Donor  0.0203    0.0625       3.08     24               0.462
12 Turkey         Donor  0.0168    0.0496       2.96     26               0.5
13 France         Donor  0.0187    0.0128       0.684    42               0.808
14 Switzerland    Donor  0.0195    0.00387      0.199    48               0.923
# ℹ 1 more variable: z_score <dbl>
```

Finding difference in means with pre-treatment values- this is our main result

```
scm_data_OECD_HIC_UMC %>%
    # Extract values from balance table
    grab_balance_table() %>%
    # Get differences between actual UK and synthetic and compare to donor differences
```

```
mutate(
    synth_diff = `United Kingdom` - `synthetic_United Kingdom`,
    donor_diff = `United Kingdom` - donor_sample,
    year = as.integer(str_extract(variable, "\\d{4}"))
) %>%
# Make into a tidy data format
pivot_longer(
    cols = c(synth_diff, donor_diff),
    names_to = "type",
    values_to = "diff"
) %>%
# Recode values to match those in paper result
mutate(type = recode(type, "synth_diff" = "Synthetic control", "donor_diff" = "Unweighted sam
# Plot
ggplot(aes(x = diff, y = factor(year), color = type)) +
geom_point(size = 3) +
geom_vline(xintercept = 0, linetype = "dashed") +
# Match colors used by paper figure
scale_color_manual(values = c(
    "Synthetic control" = "#E07B6A",
    "Unweighted sample" = "#5BBCD6"
)) +
labs(
    x = "Difference in means with pre-treatment values in the UK",
    y = "Year",
    # Take out legend title the lazy way
    color = "",
    title = "Balance in weighted synthetic control and in unweighted sample"
) +
# Get all the lines out
theme_classic()
```


Balance in weighted synthetic control and in unweighted sample
Year
2000
1999
1998
1997
1996
1995
1994
1993
1992
1991
1990
1989
1988
1987
1986
1985
1984
1983
1982
1981
1980
-1.0
-0.5
0.0
0.5
1.0
1.5
Difference in means with pre-treatment values in the UK
Synthetic control
Unweighted sample